

# P versus NP — A Public Statement

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<https://github.com/RayFromBoston/Public-Research>

*This is a public statement. It is not the technical paper. The technical paper, the programs, and the proofs stay off the public web. That is deliberate.*

The Clay Mathematics Institute asked, in 2000, whether every yes that is cheap to check is also cheap to find. That is P versus NP. Cook wrote it as one tick for every language: either finding is always as cheap as checking, or it is not.

That tick is the wrong object.

They treat “the problem” as something that lives in a machine encoding: a checker that accepts a short witness, and a search for that witness. Easy to check a seating chart against a conflict list. Hard to generate the chart by trying combinations. That example is theirs. It sounds like two different jobs. It is one job, held two ways. Checking is the rule used as “is this right?” Finding is the same rule, walked from the question to the answer. When the rule is complete, those two acts are the same act. When the rule is missing, the gap is the missing rule. It is not a fact about the problem. It is a fact about the representation.

Tick “always cheap” for every language and you have claimed that a checker with missing rules already walks. It does not. Tick “always expensive” for every language and you have claimed that even a complete rule still has a hunt. It does not. The dilemma is a classification of one blank encoding. It is not a classification of problems.

The right question is: for this checker, are the rules complete? If they are, finding and checking cost the same. If they are not, the expense is what is missing, and the work is to recover the rule — not to search a witness space.

We solved that. Cognitive Systems Incorporated has the method, the mathematics, and the runs. SAT, uniqueness, unsatisfiability, and incomplete boards are four actual answers. A-or-B on the

Turing machine misses all four. The same sitting — one object, two readings, the hunt as a missing representation — is the solution, not a guess about P and NP as two boxes.

It is the same sitting on the other Clay problems we took up. Poincaré was already Perelman's. We did not redo it. The others were written as two counts, or two boxes, or a hunt. They are not. They are one object, read two ways.

### **Navier–Stokes**

Clay asked for one tick, all fluids: always smooth, or a blow-up. Water is not glycerol. Air has a mean free path. Superfluid helium is not in their viscosity box. Smooth or break is the material at a length, not a field copied to every scale.

### **Yang–Mills**

Clay asked for a mass gap on a simple compact group. The field strength has two terms. Electromagnetism drops the self-coupling term and keeps a massless photon. A simple group keeps both terms. A free massless gluon does not persist. The gap is that fact, not a second theory.

### **Riemann**

Clay asked you to hunt zeros in a strip. Unique factorization already closed the integers. The completed zeta is that closure, read two ways. The critical line is where those two windows sit at one location.

### **Birch and Swinnerton-Dyer**

Clay asked whether two integers match: the rank of rational points, and the order of vanishing of  $L$ . They are one product, two readings. The rank is the exponent of that product.

### **Hodge**

Clay asked whether Hodge classes and algebraic cycles are two counts that luckily match. They are one variety: nested equations, and the footprint of that nest. The span is that footprint, over the rationals.

Same premise. Same error in the official sentence. Same kind of solution. Different fields. We are not asking the public to take this on trust as a slogan. We are stating that the work exists, that it has been written as full papers with the mathematics, and that it is not going on the open internet.

## **Why this is not posted**

It is not responsible to post a working solution to P versus NP where anyone can copy it. Hardness, as they currently use it, sits under a great deal of cryptography, logistics, and national infrastructure. Publishing the method in full would not be a courtesy to mathematics. It would be a gift to every actor who wanted the other side of those locks. We will not do that.

If you are a defense contractor, a U.S. Government research lab, or a cleared partner with a legitimate need to verify the claim, write to [Ray@CogSysAi.com](mailto:Ray@CogSysAi.com). Cognitive Systems Incorporated will show you the mathematics, the programs, and the runs in a setting that is responsible. We will prove it. We will not post it.

The public record is this statement, the named problems, and the invitation. The proofs stay with us until the right people ask.

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